

Vishesh Prasad

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AREAS OF INTEREST

Applied machine learning for real-world systems, reinforcement learning, adversarial ML & robustness, generative modeling, multi-agent systems, game theory, computer vision, robotics, and problem solving.

EDUCATION

- **University of Illinois Urbana-Champaign** August 2025 - Expected: December 2026
M.S. Electrical & Computer Engineering, Specialization: Machine Learning Urbana, IL
 - Thesis Advisor: Dr. Arindam Banerjee
 - Academic Advisor: Dr. Huan Zhang
- **University of Illinois Urbana-Champaign** August 2021 - May 2025
B.S. Computer Engineering Urbana, IL
 - Highest Honors, James Scholar, Dean's List
 - Minors: Mathematics, Statistics, Econometrics
 - Senior thesis: Extractive Summarization for Mathematically Rich Texts

EXPERIENCE

- **Research Assistant** October 2025 - Present
CS at University of Illinois Urbana-Champaign
 - In-Progress Projects:
 - * Robustness of generative models for offline reinforcement learning
 - * Adversarial reinforcement learning
 - * General preference multi-turn alignment for LLM agents
 - Advisor: Dr. Arindam Banerjee
- **Machine Learning Engineering Intern** May 2026 - August 2026
Molex Fremont, CA
 - Developed deep learning models for computer vision and image processing to automate optical circuit switch monitoring, modernizing a legacy process too slow and inaccurate to scale
 - Engineered and deployed a production-grade ML pipeline, reducing processing time by 98.33%
 - Advisor: Dr. Grant Zheng
- **Research Assistant** August 2023 - May 2026
Mathematical Language Processing Group (MLP), ECE at University of Illinois Urbana-Champaign 
 - Designed novel algorithms and LLM techniques for mathematical relation extraction (Preprint [\[M.1\]](#))
 - Advisor: Dr. Nickvash Kani
- **Research Assistant** August 2024 - May 2025
IBM-Illinois Discovery Accelerator Institute (IIDAI) 
 - Run-time and accuracy optimization of machine learning applications with a novel ML compiler
 - Advisors: Dr. Sasa Misailovic, Dr. Vikram Adve
- **Research Assistant** June 2024 - December 2024
Software Engineering and Analysis Lab (SEAL), CS at Cornell University
 - Machine Learning operations (MLOps) research study looking at MLflow, Metaflow, BentoML, ZenML, and Kubeflow
 - Advisor: Dr. Saikat Dutta
- **Software Engineering Intern** June 2024 - August 2024
GEICO Chevy Chase, MD
 - Developed a knowledge graph in Neo4j modeling thousands of cloud and on-prem entities and their dependencies to support cloud-to-on-prem migration planning
 - Extracted and modeled entity relationships to enable downstream cost-optimization analysis, infrastructure placement decisions, and ML based predictions
 - Built a Node.js application for performance-optimized graph querying and visualization, enabling engineers to explore infrastructure topology and dependencies

• **Embedded Software Developer Intern**

May 2023 - December 2023

Lumentum

Dallas, TX

- Extended a C# GUI's device control and testing capabilities for 100G optical transceiver module development by modifying device drivers to enable lower-level register access
- Set up IAR toolchains for ARM with Jenkins CI integration and debugged embedded C firmware
- Diagnosed field failures by replicating customer-reported issues through microphonic testing

TEACHING EXPERIENCE AND MENTORSHIP

• **Graduate Teaching Assistant**

August 2025 - Present

University of Illinois Department of Electrical and Computer Engineering

- ECE 484 – Principles of Safe Autonomy (Aug. 2026 - Present)
- Head TA: CS 446/ECE 449 – Machine Learning (Jan. 2026 - May. 2026)
- CS/ECE 374 – Algorithms & Models of Computation (Aug. 2025 - Dec. 2025)

• **URSA Graduate Mentor**

January 2026 - May 2026

Undergraduate Research in Scientific Advancement, Grainger College of Engineering at UIUC



- Mentored 4 undergraduates new to research on a full research project

• **Undergraduate Teaching Assistant**

August 2022 - May 2025

University of Illinois Department of Electrical and Computer Engineering

- ECE 364 – Programming Methods for Machine Learning (Jan. 2025 - May 2025)
- CS/ECE 374 – Algorithms & Models of Computation (Jan. 2024 - Dec. 2024)
- Engineering Economics – Course Development as Head TA (Apr. 2024 - July 2024)
- ECE 110 – Introduction to Electronics (Aug. 2022 - Aug. 2023)

TECHNICAL SKILLS

- **Languages:** Python, C++, CUDA, x86 Assembly, R, OCaml, Golang, C, C#, JavaScript
- **ML and Tools:** Pandas, NumPy, PyTorch, D4RL, OpenAI Gym, OpenCV, ONNX, ROS, Gazebo

RELEVANT COURSEWORK

- **Graduate Coursework:** Statistical Reinforcement Learning, Sequential Decision Making, Deep Generative Models, Algorithmic Game Theory, Statistical Detection and Estimation Theory, Statistical Learning Theory, Random Processes, Algorithms, Algorithmic Market Microstructure
- **Upper-level Coursework:** Machine Learning, Digital Signal Processing, Stochastic Processes, Real Analysis, Financial Econometrics, Optimization, Economic Game Theory, Probabilistic Graph Models, Parallel Programming, Computer Systems, Logic Synthesis

HONORS AND AWARDS

• **Harold L. Olesen Award for Excellence in Teaching**

December 2025

Department of Electrical and Computer Engineering at Illinois



• **Edward C. Jordan Award**

April 2025

Department of Electrical and Computer Engineering at Illinois



• **Indira Gunda Saladi Research Scholarship**

April 2025

Department of Electrical and Computer Engineering at Illinois



• **Illinois Scholars Undergraduate Research Scholarship**

May 2024

Illinois Scholars Undergraduate Research (ISUR) Program



• **Edmund J. James Scholar**

August 2021 - May 2025

Department of Electrical and Computer Engineering at Illinois

• **Dean's List**

University of Illinois Urbana-Champaign

PUBLICATIONS

M=MANUSCRIPT(PRE-PRINT)

- [M.1] Prasad, et al. (2026). **Mathematical Derivation Graphs: A Relation Extraction Task in STEM Manuscripts**. In *arXiv*.

STUDENT ORGANIZATIONS

• **Association for Quantitative Trading Education (AQTE)**

April 2022 - January 2023

- Education Director

• **EntreCOPRS - Startup Consulting**

February 2022 - May 2023

- Technology Director, Strategy Manager, Senior Consultant, Consultant

ADDITIONAL INFORMATION

Interests: Chess, Hiking, Reading, ... Page 2 of 2 ..., Tenor Saxophone, Pottery